Houthakker generates a function of the variable inputs only, and it exhibits diminishing returns. We also have some differences in the way we model the heterogeneity across production units: Houthakker's production units differ in terms of their static capacity constraints (interpreted as the quantities of the unmodelled fixed production factor), whereas mine differ due to random productivities that evolve over time. But despite all these differences, note that the factor allocation rule in the search equilibrium, (8), is reminiscent of (26) and that (18) and (19) are the analogues of (27) and (28).

Another technical difference between both analyses is that by virtue of working within a partial equilibrium framework, Houthakker had the freedom to assume the particular shape of the heterogeneity across production units to be used in the integrals on the R.H.S. of (27) and (28). Loosely speaking, his production units could not "move around the distribution". In contrast, because I am working within the context of a dynamic equilibrium model, I do not have the freedom to choose the distribution that shows up in the aggregates. Specifically, I cannot choose H, which corresponds to Houthakker's φ, because H is not a primitive. I can only specify G. This introduces an additional layer of complexity, since the mapping between the primitive distribution of shocks, G, and the equilibrium distribution, H, depends on equilibrium forces.[24]

From the perspective of understanding the determinants of TFP, a crucial difference is that the shift parameter in Houthakker's production function (what he called C) is solely a function of the parameters in the primitive productivity distribution. But in the model I developed above, government policy can—through the agents' decisions—affect the multiplicative factor in front of the aggregate production function.

At this point it may be useful to stress that in this context, by an "aggregate production function" I mean *an equilibrium relationship between measured aggregate inputs and output*. In terms of (22), the fact that this is an *equilibrium* relationship implies that in general, a change in policy parameters (*e.g.* unemployment benefits) will not affect γ but will affect N, K_e, and A. Intuitively, one may regard A as an additional "factor", which has to do with how production is organized across micro-units, and naturally this factor responds to policy, much as, say, hours worked would. The contemporary reader may wonder: if this is what an aggregate production function is, then what does "aggregation" mean?

The classic problem of aggregation of production functions consists of finding an aggregate relationship between an output aggregate and input aggregates. For example, suppose the production function for the j-th productive unit is $y_j = f^j(K_j, N_j)$. Assume capital and labour are homogeneous, and let $K = \sum_j K_j$ and $N = \sum_j N_j$. The problem of aggregation consists of finding conditions under which there exists a function F such that $\sum_j y_j = F(K, N)$. It has been known since the late 1940's that if nothing else is said, the general conditions under which such aggregation can be performed are extremely restrictive. Except in some very special cases, in an economy with multiple production units, the amount of total output that can be obtained from different levels of total inputs is not unique; the distribution of the inputs matters.[25] For this reason, the aggregation literature has traditionally defined an aggregate production function as a relationship between the amount of total output produced, not from any arbitrary levels of total inputs, but for levels of total inputs that satisfy some conditions. For example, the aggregate production function in Houthakker (1955–1956) is a relationship between aggregate output and aggregate inputs—given that the allocation of inputs across production units is consistent with

24. In the model of Section 2, with independently and identically distributed productivity shocks, H is simply the truncation of G. Since the Pareto family is closed under truncations, the mapping between the primitive G and the equilibrium distribution H is simple, as is evident from (20) and (21). This simplicity is retained in the formulation with serially correlated shocks that I work out in Section 8.1. The mapping between G and H is less straightforward in other cases; for example, see the results in Section 8.2.

25. To see this point, try to add up two identical Cobb–Douglas production functions without further restrictions. For more on this, see Fisher (1969) and references therein.

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profit maximization. Similarly, the aggregate production function (22) is a relationship between aggregate output and aggregate inputs—given that the allocation of inputs across production units is consistent with equilibrium.[26]

6.2. *Measurement*

The aggregate production function in (22) expresses output as a function of the aggregate number of hours worked, N, and the total amount of capital hired by firms with filled jobs, K_e. The aggregate is Cobb–Douglas despite fixed proportions in the micro-level technologies. This result is obtained when there is hoarding and capital utilization is imperfectly measured, that is, when the capital stock included as an argument in the aggregate production function is the one in place in firms that are employing workers, although some of it may be idle. Notice that if there is no hoarding in equilibrium (*i.e.* if $\mu = R$), then $N = K_e$, and $F(K_e, N) = AK_e^{\gamma} N^{1-\gamma} = AK_e$. If there is hoarding but utilization is perfectly measured, then aggregate output is again linear in the relevant capital stock. To see this explicitly, let K_p denote the capital stock being used in production, that is, $K_p = [1 - H(\mu)]K_e$. Then it follows from (18) that $Y = A''K_p$, with $A'' \equiv E_H(x \mid x \geq \phi)$. So the imperfect measurement of capital utilization as described, coupled with some degree of hoarding, causes the aggregate to look Cobb–Douglas in capital and hours despite fixed proportions in the micro-production functions. Since having firm–worker pairs that sometimes choose to be inactive affects the shape of the aggregates, in Appendix B, I derive conditions on primitives such that the equilibrium will exhibit this labour-hoarding property.

Although hoarding and the measurement of factor utilization have implications for the shape of the aggregate production function, one should not conclude that the theory predicts only productivity differentials due to hoarding and mismeasured utilization. For example, even if there is no hoarding at all, the TFP term is equal to $\alpha R/(\alpha - 1)$, the mean of the productivity distribution of active firms. Of course, the way inputs are measured in general, and in particular the extent to which their utilization rates are correctly measured, *will* affect the TFP factor when it is measured as a residual. I turn to this issue next. Consider how the observed level of TFP is affected by the different ways of measuring aggregate inputs that can be found in the growth accounting literature. The measure of capital input used by Hall and Jones (1999), for example, did not adjust for utilization. This means that something like K instead of K_e was used as an input in the aggregate production function, implying $\hat{F}(K, N) = \hat{A}K^{\gamma} N^{1-\gamma}$, with $\hat{A} = \left[\frac{1-u}{1-(1-\theta)u}\right]^{\gamma} A$. In addition, Hall and Jones (1999) report they did not have data on hours per worker for all countries in their sample, so they used the number of employed workers instead of hours worked as a measure of labour input. Letting $E = 1 - u$ denote employment and using (17), the number of hours worked is $N = (R/\mu)^{1/\gamma} \frac{KE}{1-(1-\theta)u}$, so—from the standpoint of the theory developed here—the aggregate relationship between inputs, output, and TFP that they measured was $\tilde{F}(K, E) = \tilde{A}K^{\gamma} E^{1-\gamma}$, with $\tilde{A} = \left[\frac{(R/\mu)^{1/\gamma} K}{1-(1-\theta)u}\right]^{1-\gamma} \hat{A}$. In Section 7, I will quantify the impact that each of the labour-market policies introduced in Section 5 has on the TFP residual for these different ways of measuring factor inputs.

6.3. *Standard growth accounting*

Macroeconomists interested in identifying the sources of differences in output per worker find it useful to relate aggregate measures of output to aggregate measures of capital and labour through an aggregate production function. As a matter of accounting, this abstraction allows one to

26. These notions should be familiar: The standard aggregation result for the one-sector growth model with constant-returns technologies also relies on a distribution of inputs consistent with firm profit maximization.

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attribute the observed variation in output levels to differences in the quantities of the inputs and to differences in an unexplained TFP residual. The overwhelming majority of these growth accounting exercises adopts a Cobb–Douglas specification for the aggregate production function.[27]

Virtually all these growth accounting exercises rest either explicitly or implicitly on the premise that the model with frictionless markets and competitive pricing is a good approximation to the actual economy. But what if it is not? What if the actual economy lay closer to a model where labour markets are frictional and prices determined by bargaining? Would adopting this alternative view—as many economists have over the last 20 years to study various macro questions—imply that the standard growth accounting exercises that adopt a Cobb–Douglas specification, for example, Klenow and Rodríguez-Clare (1997), Hall and Jones (1999), Parente and Prescott (2000), and many others, are misspecified in ways that render them completely uninformative?

On a positive note, the analysis of Section 4 offers a relatively standard search model—with its underlying meeting frictions, simple fixed-proportions micro-level production technologies, and prices determined by bargaining—according to which the standard growth accounting exercises that adopt a Cobb–Douglas specification for the aggregate technology have in fact chosen the right functional form. The problem is, of course, that the direct mapping between factor income shares and the elasticities in the aggregate production function that exists in the frictionless model with competitive prices that underlies the standard growth accounting exercises is distorted in the search economy. Suppose, for example, that a growth accountant assumes $Y = BK_e^{\sigma} N^{1-\sigma}$ and works under the presumption that factor markets are competitive and frictionless, when instead the actual economy is as described in Sections 2–4, so that $Y = AK_e^{\gamma} N^{1-\gamma}$ is, in fact, the correct relationship between aggregate output and inputs. The TFP residual measured by this accountant would be $B = [1 - H(\mu)]^{\sigma-\gamma} A$.[28] Since he uses the competitive model to think about the actual economy, the accountant will be inclined to set σ equal to the capital share in income from his data. In contrast, an accountant who organizes his thoughts using the model in Section 4 will be led to look at purely technological features of the actual economy to pin down γ.[29] The relationship between the size and direction of the bias and these fundamentals is simple: the absolute size of the bias increases in the degree of hoarding. No hoarding implies the accountant with the "wrong" (competitive) model would still recover the "right" TFP factor. The direction of the bias, when there is one, depends on the sign of $\sigma - \gamma$. If there is some hoarding, and the capital share in the data is smaller than the technological parameter γ, then the accountant guided by the competitive model will overestimate the size of the true productivity residual.

In summary, the theory developed here provides foundations—which rest on a model with a frictional labour market and noncompetitive wages—for the Cobb–Douglas specification of the aggregate production function commonly used by macroeconomists for growth accounting. The theory implies a mapping between observables and the elasticity parameter in the aggregate that is different from that implied by the competitive model, and it can be used to assess the implications of this difference for the size of the measured residual.

27. The most common justification for adopting this particular specification is that, at least for some countries, notably the U.S. and the U.K., factor income shares have remained roughly constant over long periods of time, despite rather large secular changes in factor prices. (See figure 1 in Gollin, 2002, and the bottom panel of figure 1 in Blanchard, 1997.) And, as pointed out by the eponym Cobb and Douglas (1928), the specification implies constant factor shares of income if factors are paid their marginal products.

28. Here, I have assumed that the accountant measures only the capital in place in firms that have positive employment, K_e, but it may be more realistic to suppose he instead measures the total capital stock, K. If this is the case, the measured residual would be $B = \left[\frac{1-u}{1-(1-\theta)u}\right]^{\sigma} [1 - H(\mu)]^{\sigma-\gamma} A$.

29. Recall that $\gamma = 1/\alpha$ is a parameter of the underlying distribution of shocks. So, for example, the growth accountant may attempt to infer it from the distribution of value added across production units. I thank Esteban Rossi-Hansberg for suggesting this possibility.

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The emphasis throughout the paper is on understanding the determinants of the TFP residual, and factor income shares are not central to this issue from the standpoint of the theory. However, factor income shares play a crucial role in standard growth accounting, so the reader may feel that some remarks on the nature of the factor shares in the model may be in order. In this model, factor income shares are not immutable technological constants: they depend not only on technological considerations but also on labour-market policies or anything else that affects the division of rents between workers and employers. For example, policies that increase the unemployment rate will tend to decrease the labour share.[30]

I conclude this section by commenting on where this paper stands relative to the literature that uses growth accounting to frame some fundamental questions in development economics. In the model developed here, differences in TFP arise from composition effects, that is, through the way production inputs, particularly hours of work, are allocated across micro-production units. This feature of the theory naturally leads one to think of institutions that may distort this allocation of resources, and this is where labour-market policies come in as a natural suspect. Note that the mechanisms at work in this paper allow two economies to exhibit different levels of TFP even if production units in both have access to the same technology and are subject to identical shocks. But the world's richest and poorest countries seem to have very different production technologies in place, so trying to explain 30-fold output-per-worker differences, keeping technologies the same and relying exclusively on policies that only distort the allocation of resources (labour or otherwise), would seem misguided.[31] In the following section I calibrate the model economy and show that—if assessed within the context of a homogeneous set of countries, for example, the OECD—the magnitude of the effects of labour-market policies on the level of measured TFP can in some cases be significant.

30. Blanchard (1997) documents the decline in the labour share for continental European countries (France, Germany, Italy, and Spain) during 1970–1995. He offers changes in policies and institutional considerations that affected the relative bargaining positions of employers and workers as plausible explanations. Interestingly—from the point of view of the competitive model—non-constant factor shares imply that the aggregate production function could not be Cobb–Douglas. But from the point of view of the aggregative model developed here, this conclusion does not follow: it can exhibit non-constant factor shares and, with a Pareto distribution of idiosyncratic productivity shocks, it can still aggregate up to a Cobb–Douglas production function. To illustrate, let s_w and s_k denote the labour and capital shares, respectively, and consider a version of the basic model in which ϕ indexes a worker's disutility from work (see Appendix B), $b = \tau_b k$, and there is no fixed cost (*i.e.* $C = 0$). Then,

$$s_w = \beta + \xi(1-\beta) - z_k \frac{K_e}{Y} + z_n \frac{N}{Y} \quad \text{and} \quad s_k = 1 - s_w,$$

where $z_k \equiv [1-(1-\xi)(1-\beta)]c - (1-\xi)[\beta c\theta + (1-\beta)\tau_b]$, $z_n \equiv (1-\xi)(1-\beta)\phi$, and ξ is an accounting parameter that denotes the fraction of the firm's rents imputed as labour income. So, for example, if $\xi = 1$, then $s_k = \frac{cK_e}{Y} = \frac{c}{F_1(K_e,N)}\gamma$.

31. In order to make the quantitative leap to development accounting within the context of this model, one may have to formalize the notion that the policies considered not only distort the allocation of inputs but also hinder the adoption of better technologies. Although they focus on monopoly rights in their formal modelling of barriers to technology adoption, Parente and Prescott (2000) mention several labour-market policies as examples of "barriers to riches": "In India, for example, firms with more than 100 workers must obtain the government's permission to terminate any worker, and firms of all sizes are subject to state certification of changes in the tasks associated with a job" (pp. 107–108). "Another way the state protects the monopoly rights is by requiring large severance payments to laid-off workers". "Also in India, regulations require certain firms to award workers with lifetime employment and require firms with more than 25 workers to use official labour exchanges to fill any vacancy" (p. 108). "In Bangladesh, for example, private buyers of the state-owned jute mills were prohibited for one year from laying off any of the workforce they inherited. After one year, a worker could be laid off but not without a large severance payment". Parente and Prescott use these examples as instances of policies that can lower TFP by making technology adoption costly, whereas I emphasize that these policies have a direct impact on the level of TFP through composition effects—even for given technology. See footnote 1 for some issues that the model developed here may be well suited for.

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